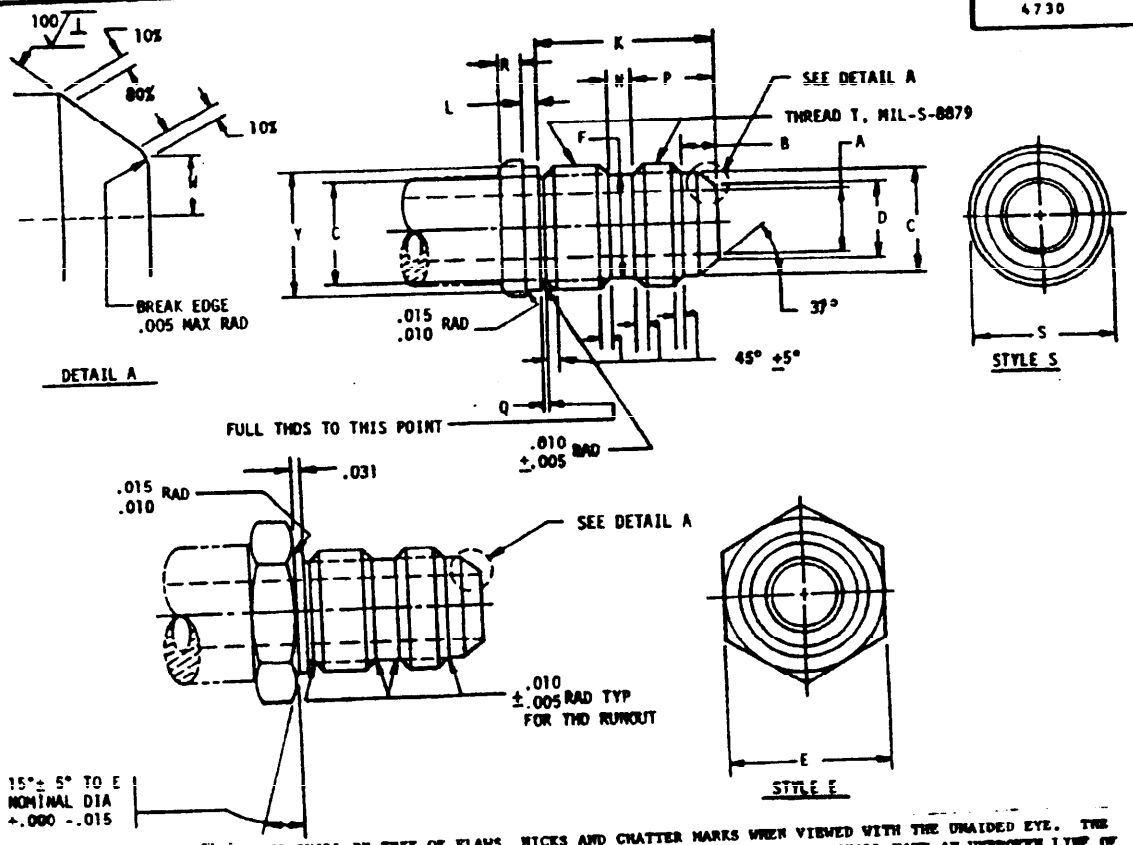
FED. SUP CLASS
4730



(G) THE CONICAL SEAL SURFACE SHALL BE FREE OF FLAWS, NICKS AND CHATTER MARKS WHEN VIEWED WITH THE UNAIDED EYE. THE CENTER 80% OF THE CONICAL SURFACE (SEE DETAIL A) SHALL ALSO HAVE NO LOW SPOTS AND SHALL HAVE AN UNBROKEN LINE OF CONTACT TO SHOW ALL AROUND WHEN ROTATED IN CONTACT 74° 0' ± 1' INCLUDED ANGLE CAGE LIGHTLY COATED BY RUBBING WITH CARBON PAPER OR BY AN EQUIVALENT BLUEING METHOD.

L AS LISTED PROVIDES FOR A WASHER TO FORM BACKING FACE AGAINST BULKHEAD PLUS 1/32 LAND FOR POSITIONING IN BULKHEAD. WHEN DESIGN OF FITTING INCLUDES A HEX OR FACE EQUAL TO OR GREATER THAN E, NO WASHER IS REQUIRED AND THIS DIMENSION WILL BE .031 AS IN STYLE E.

F SHALL BE 100AA MAX, FREE FROM BURRS, HAVING NO LONGITUDINAL OR SPIRAL TOOL MARKS. ANNULAR TOOL MARKS PERMISSIBLE TO 32AA MAX. TOLERANCE ON F MAY BE +.002 -.003 PROVIDED NO THREAD MARKS ARE PRESENT.

SQUARENESS BETWEEN THREAD T (PITCH DIA) AND FACE OF HEX (STYLE E) OR BACKING FACE (STYLE S) SHALL BE WITHIN H.

EXAMPLE OF DESIGN IDENTIFICATION NUMBER: MS33657[16]

(G)
16 = SIZE OF FITTING END IN 16THS.
E = STYLE E
S = STYLE S
BASIC DESIGN NUMBER

BREAK ALL SHARP EDGES AND REMOVE ALL HANGING BURRS AND SLIVERS.

DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED, TOLERANCES:
ANGLES ± 1/2°, DECIMALS ± .016.

DIMENSIONS D,C,F AND T (PITCH DIA) SHALL BE CONCENTRIC TO EACH OTHER AND WITH THE 37° ANGLE, MEASURED AND CALCULATED AS AN INCLUDED ANGLE, WITHIN .005 TOTAL INDICATOR READING. OTHER DIAMETERS SHALL BE CONCENTRIC WITHIN .010 TIR.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS.

SURFACE ROUGHNESS IN ACCORDANCE WITH ANSI B46.1.

REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

FITTINGS WITH MIL-S-7742 THREADS OF THE SAME SIZE, PITCH AND TOLERANCE CLASS MAY BE USED UNTIL STOCKS ARE EXHAUSTED, BUT ARE NOT TO BE SUPPLIED AFTER 1 JAN 1976.

FOR CHANGE SEE SHEET 1. THIS IS A DESIGN STANDARD. NOT TO BE USED AS A PART NUMBER.

P.A. AIR FORCE - 82 Other Cust ARMY - AR NAVY - AS AIR FORCE - 99	INTERNATIONAL INTEREST	TITLE FITTING END, STANDARD DIMENSIONS FOR BULKHEAD FLARED TUBE CONNECTION.	MILITARY STANDARD MS 33657 SHEET 1 OF 2
PROCUREMENT SPECIFICATION NONE	SUPERSEDES AND10057		

4 730-020 7

DD FORM 1 MAY 73 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

APPROVED 15 MAY 57 REVISED (E) 11 OCT 74 (F) 15 MAY 75 (G) 10 [illegible] 85

USER ACTIVITIES ARMY - ME

REVIEWER ACTIVITIES DLA - CS ARMY - MI

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Review activities: DLA-CS
ARMY-MI

User activities: ARMY-ME

P.A. AIR FORCE- 82

Other Cust
ARMY-AR
NAVY-AS
AIR FORCE-99

PROCUREMENT SPECIFICATION
NONE

SUPERSEDES:
AMD10057

TITLE
FITTING END, STANDARD DIMENSIONS FOR BULKHEAD FLARED TUBE CONNECTION

MILITARY STANDARD

MS33657

SHEET 2 OF 2

FED. SUP CLASS
4730

DD FORM 672-1 (Reverse)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

THIS IS A DESIGN STANDARD, NOT TO BE USED AS A PART NUMBER

TABLE I DIMENSIONS

SIZE NO	TUBING OD	THREAD T MIL-S-8879	A DIA		B +.015 -.000	C +.000 -.005 DIA	D ±.003 DIA	E		F +.000 -.003 DIA	H MAX	K	L	N +.010 -.000	P ±.005	Q +.015 -.000	R	S DIA	W MIN	Y MAX	Y MIN
2	1/8	5/16-24 UNJF-3A	.062	+.003 -.004	.177	.245	.083	.563	+.003 -.004	.250	.005	.922	.094	.125	.385	.063	.109	.438	.005	.3125	.3053
3	3/16	3/8 -24 UNJF-3A	.125			.307	.146	.625		.312				.131				.500		.3750	.3678
4	1/4	7/16-20 UNJF-3A	.172		.193	.359	.193	.688		.364		1.016		.140	.443	.075	.125	.562		.4375	.4294
5	5/16	1/2 -20 UNJF-3A	.234			.421	.255	.750		.426								.625		.5000	.4919
6	3/8	9/16-18 UNJF-3A	.297	±.004	.198	.476	.318	.813		.481		1.094		.156	.476	.083	.141	.688		.5625	.5538
8	1/2	3/4 -16 UNJF-3A	.391		.253	.654	.426	1.000		.660		1.250		.187	.565	.094	.156	.875	.012	.7500	.7406
10	5/8	7/8 -14 UNJF-3A	.484		.266	.767	.539	1.125		.773		1.391		.219	.623	.107	.172	1.000	.022	.8750	.8647
12	3/4	1-1/16-12 UNJ-3A	.609	±.005	.315	.938	.664	1.375		.945	.008	1.562	.125	.234	.732	.125	.188	1.188	.029	1.0625	1.0511
16	1	1-5/16-12 UNJ-3A	.844	+.007 -.005		1.188	.913	1.625	±.016	1.195								1.438		1.3125	1.3011
20	1-1/4	1-5/8 -12 UNJ-3A	1.078	+.008 -.005	.367	1.501	1.147	1.875		1.507		1.609			.784			1.750		1.6250	1.6136
24	1-1/2	1-7/8 -12 UNJ-3A	1.312		.378	1.750	1.381	2.125		1.756		1.625			.795			2.000		1.8750	1.8636
28	1-3/4	2-1/4-12UNJ-3A	1.547	+.010 -.005	.451	2.125	1.646	2.500	±.020	2.131		1.781			.868			2.375	.044	2.2500	2.2386
32	2	2-1/2 -12 UNJ-3A	1.781		.461	2.375	1.880	2.750		2.381		1.906			.878			2.625		2.5000	2.4886
40	2-1/2	3 -12 UNJ-3A	2.281		.395	2.882	2.380	3.250		2.881		1.719			.895			3.125		3.0000	2.9886
48	3	3-1/2 -12 UNJ-3A	2.781			3.382	2.880	3.750		3.381		1.812			.978			3.625		3.5000	3.4886

APPROVED 15 MAY 57 REVISED

(G) FOR CHANGE SEE SHEET 1.

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